ThermoTwin: status and plan

Team Alebaple · CODE APEX 3.0, Track 4 (UST) · T4-1 Healthcare cold chain · Status as of 21 October 2026 · Grand finale 24 October 2026

In one paragraph. ThermoTwin is a software early-warning layer for vaccine fridges. It reads the temperature that existing loggers (eVIN) already record, estimates the vaccine's own temperature with a per-fridge digital twin (the Virtual Vial), predicts freezing 1–2 hours ahead with a LightGBM risk model, and tells staff the likely cause and what to do. WHO alarms keep running unchanged. Research, PRD, pitch deck, a clickable demo UI and a working core system are done. On simulated data, ThermoTwin warns a median of **90 minutes** before the vaccine reaches 0 °C, where the WHO rule fires **30 minutes after**, with **0.48** false alerts per fridge-month. The next three days connect the model to the UI, add the heat and cause models, and validate.

41 / 41	90 min	0.48	0.08 °C
freeze events caught on held-out fridges	median warning before vaccine hits 0 °C	false alerts per fridge-month (budget 1.0)	Virtual Vial error vs true vaccine temp (raw air: 0.46 °C)

All results are from the ColdChain Gym simulator, not field data.

1. The problem

Vaccines must stay at +2 to +8 °C. Freeze-sensitive vaccines (pentavalent, hepatitis B, DTP-containing, PCV, IPV) are permanently damaged at or below about −0.5 °C, and a frozen vial looks normal. Indian studies found vaccines below 0 °C for roughly 10–15% of storage time. Today's WHO alarms fire only after damage starts (10 h above +8 °C, or 1 h at or below −0.5 °C). India's ~25,000 eVIN loggers across about 29,000 cold-chain points (UNDP) already record every fridge, but the data is used for stock, not prediction.

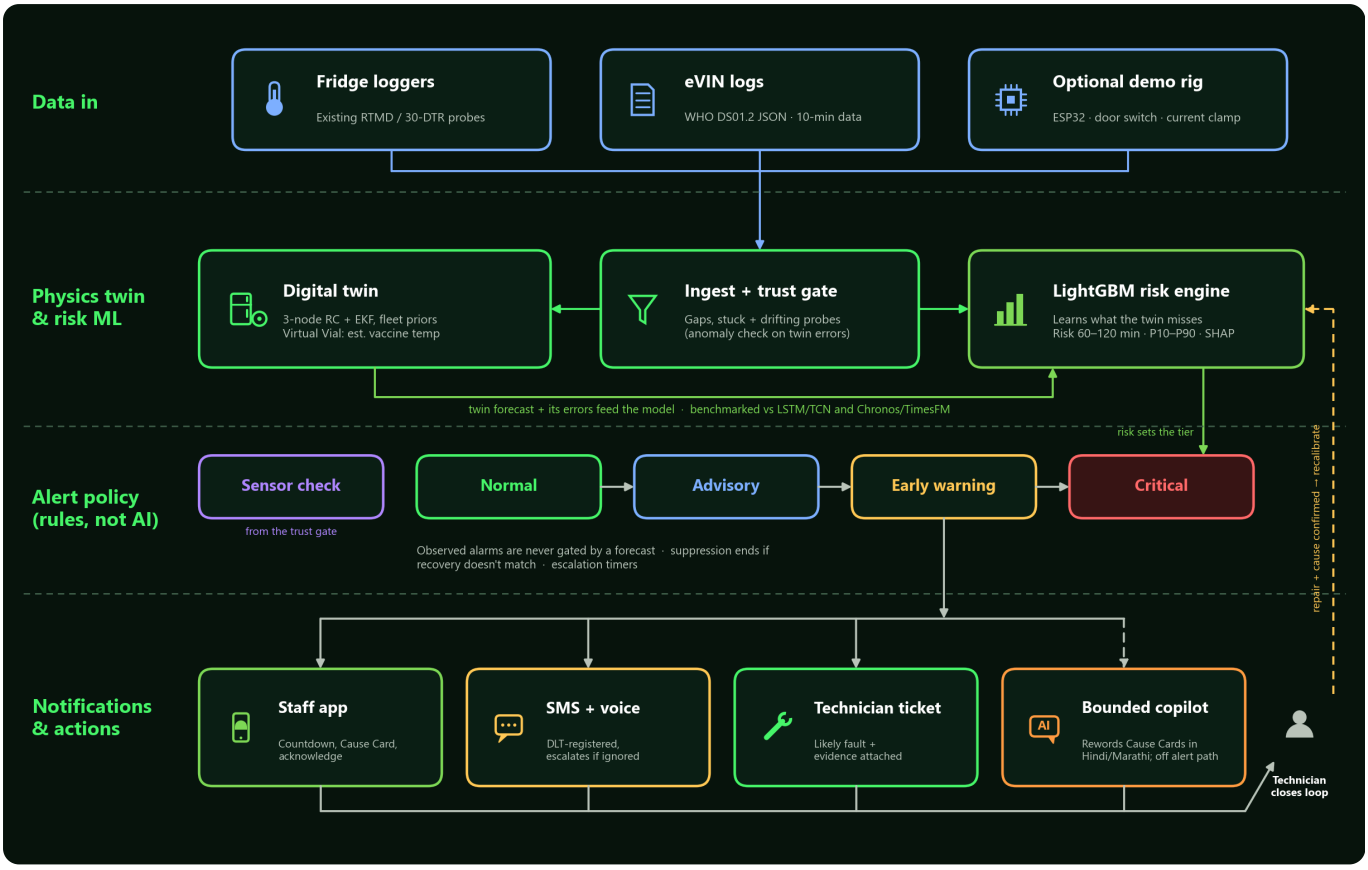
Ways cooling goes out of control: **too cold** (thermostat set or drifting cold, relay stuck on, ice-lined cold spots, overshoot after a power cut); **too warm** (power cut, door left open, warm stock, heat waves); **slow decline** (compressor wear, refrigerant leak, worn gasket, dusty condenser, ice build-up); and **a lying sensor** (stuck, drifting, dropouts).

2. What has been done

Workstream	Delivered	Status
Problem selection	Analysis of all 11 problem statements across the three packs; pitch playbook; T4-1 chosen.	**Done**
Research	Deep-research report: WHO PQS / eVIN standards, market (Nexleaf, Berlinger, Haier, Controlant…), physics and ML methods, public datasets, alarm-fatigue evidence, and a list of claims to avoid.	**Done**
PRD	Revision 4 (Claude Doc, Word export): requirements, alert policy, ML spec, design decisions, scientific validity checks, hardware section, training-data plan, model comparison.	**Done**
Pitch deck	8-slide Canva deck on the CODE APEX template; submitted for Round 1 (11 Oct) and since updated: swimlane architecture, features with digital-twin and Virtual Vial panel, lifeboat routing, scope cards, honest impact wording, sourced figures.	**Done**

Demo UI	Offline clickable demo: phone app (Home, Alerts, Cause Card, Help; Hindi toggle) and district website (Overview, Facilities, Alerts, Maintenance, Exposure Ledger, Benchmark, Settings, ILR-1 digital twin, Copilot). Packaged as `ThermoTwin_UI_demo.zip`.	**Done**
Core system	Python package `thermotwin/`: simulator, freeze labels, Kalman twin, features, LightGBM hazard model, calibration, alert policies, Replay Benchmark; 6 passing tests; one-command run (~45 s).	**Done (freeze)**
Pitch support	Speaker script for slides 5–7, UI demo script, answers on the dataset, LightGBM, the Virtual Vial formula and its validity, vaccine-aware alerts.	**Done**
Not yet done	Heat model, cause classifier, sensor-check model, API, UI wired to the live model, LSTM/foundation baselines, realism checks on public data, demo hardware rig, vaccine-aware alerts.	**Planned (§5)**

3. Architecture



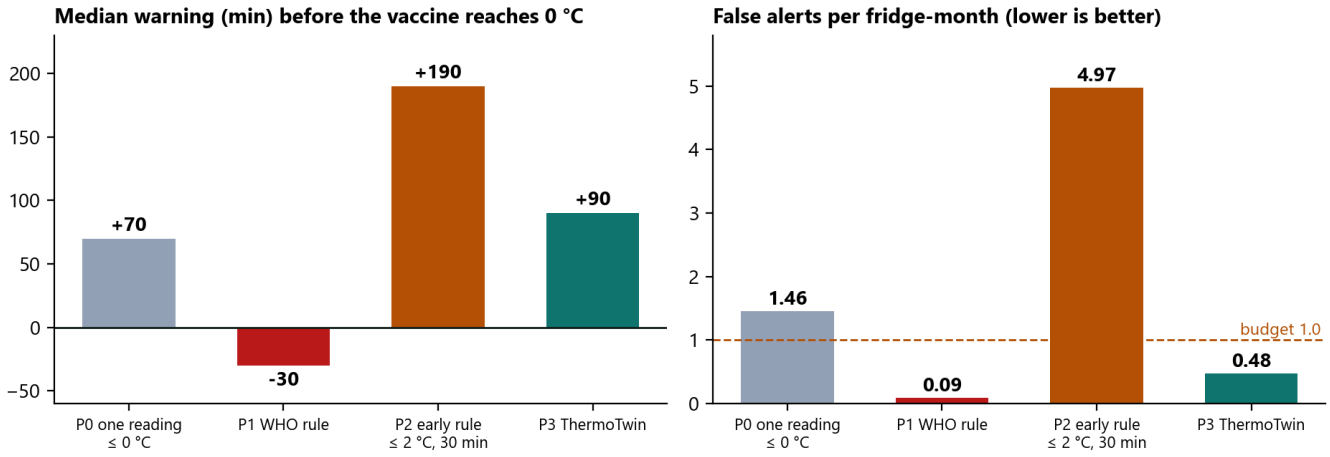
Data in (existing loggers, eVIN) → ingest and trust gate → digital twin with fleet priors (Virtual Vial) → LightGBM risk engine → rule-based alert tiers (Sensor check, Normal, Advisory, Early warning, Critical) → staff app, SMS/voice, technician ticket, bounded copilot → technician feedback and recalibration. Observed alarms are never gated by a forecast; the LLM only rewords Cause Cards.

How the core pieces work

Piece	Method
Simulator (ColdChain Gym)	3-node RC heat model (air, product, enclosure) at 1-minute steps, thermostat with hysteresis, clinic-hour door openings, Indian room temperatures, and 7 injected faults (thermostat drifting cold, relay stuck on, compressor wear, gasket leak, power cut, stuck probe, drifting probe). Logs every 10 minutes like eVIN.
Digital twin / Virtual Vial	Kalman filter per fridge, state [air, slope, vial]. The vial follows Newton's law of cooling: $dT_{vial}/dt = (T_{air} - T_{vial})/\tau$, with $\tau = mc/hA$ from the fridge-class prior. Outputs the vial estimate with an error band and 30/60/120-minute physics forecasts.
Risk model	LightGBM discrete-time hazard over twelve 10-minute slots; inputs are temperature-only features plus twin outputs. Gives calibrated 1-h and 2-h risk (isotonic), time-to-breach P10/P50/P90, and SHAP drivers for the Cause Card.
Labels	Freeze event = true vaccine temperature ≤ −0.5 °C for 60 min (from simulator ground truth only). Lead time is measured to the true vaccine crossing 0 °C.
Alert threshold	Chosen on validation fridges as the most sensitive setting within a budget of 1 false alert per fridge-month.

4. Results so far (simulated, held-out fridges)

240 fridges × 21 days (5,040 fridge-days), 178 freeze events. Split by fridge: 60% train, 20% validation, 20% test (48 fridges, 41 events).



Policy	Caught	Median warning	False alerts / fridge-month
P0 one reading ≤ 0 °C	41/41	70 min before	1.46
P1 WHO rule (≤ −0.5 °C for 60 min)	41/41	30 min after	0.09
P2 early rule (≤ 2 °C for 30 min)	41/41	190 min before	4.97
P3 ThermoTwin	**41/41**	**90 min before**	**0.48**

Reading the result. ThermoTwin warns about two hours earlier than the WHO rule, within the false-alert budget. The simple early rule warns sooner but would send about five false SMS per fridge per month, which is the alarm fatigue we are designed to avoid. The Virtual Vial tracks the true vaccine temperature with 0.08 °C mean error (raw air: 0.46 °C), and its 2-sigma band covers 91% of cases (target 95%).

Honesty notes. These are simulator results, not field results. The simulator is not yet checked against public fridge traces or a second simulator. One realism change was made: a quarter of fridges run close to 2 °C, as real clinics often do; this makes the simple rules look worse and is disclosed in the code. The deck's illustrative chart (61 min before vs 77 min after) should be replaced by these measured numbers.

5. Plan to the finale

Day	Work	Done when
21 Oct (today)	Heat-excursion model (same pipeline, heat labels). Cause classifier scored against the injected fault type. Sensor-check tier for stuck and drifting probes.	Benchmark reports freeze and heat; cause accuracy on held-out fault types.
22 Oct	FastAPI service (fridges, alerts, Cause Card, simulate-scenario). Wire the website's ILR-1 page and the phone Cause Card to the live model, replaying a simulated freeze. LSTM baseline on the same splits.	UI shows model output, not fixed numbers; LSTM row in the scorecard.
23 Oct	Realism checks against public traces (Bangkok fridges, REFIT) once licences are confirmed; cross-simulator test if the Benamirouche generator is available. Update deck slides 2 and 7 with measured numbers. Rehearse pitch and demo twice. Optional ESP32 demo rig.	Deck shows measured results; demo run-through under 3 min.
24 Oct (finale, 24 h)	Run the benchmark on any organiser data. Live fault demo (replay or rig). Stretch: vaccine-aware alerts and the lifeboat-routing calculation in code. Publish the scorecard.	Scorecard on organiser data; working live demo.

After the hackathon

- Shadow-mode pilot beside a clinic's certified logger; field calibration of τ and alert thresholds.
- Fleet priors refit per fridge, residual learning (ML on the twin's errors), Chronos/TimesFM baseline.
- Vaccine-aware alerts using eVIN stock lists and published WHO / manufacturer stability data.

6. Risks and mitigations

Risk	Mitigation
Simulator does not match real fridges	Realism gate against public traces; cross-simulator test; real data used only for testing.
Organiser data in an unknown format or temperature-only	Ingest accepts DS01.2 JSON and CSV; the model already uses temperature only.
Demo fails on stage	Offline UI with no internet dependency; recorded replay as fallback; scripted short path.
Over-claiming	Say "designed to warn earlier" until validated; never "WHO-certified", "first AI cold-chain system" or "decides vaccine safety".
Time	Core path is frozen and working; new features only after the API and UI wiring land.

7. Where everything is

Item	Location
Core system	`D:\CodeApex\thermotwin` · run `python run_core_path.py`
Demo UI	`D:\CodeApex\ui\ThermoTwin_UI_demo.zip` · double-click `START_DEMO.bat`
PRD (Revision 4)	Claude Doc "ThermoTwin PRD" · Word copy `D:\CodeApex\ThermoTwin_PRD.docx`
Pitch deck	Canva: canva.com/d/7LXEnwt4bkt6QJC
Research	`D:\CodeApex\reports\ThermoTwin cold chain market research.md`